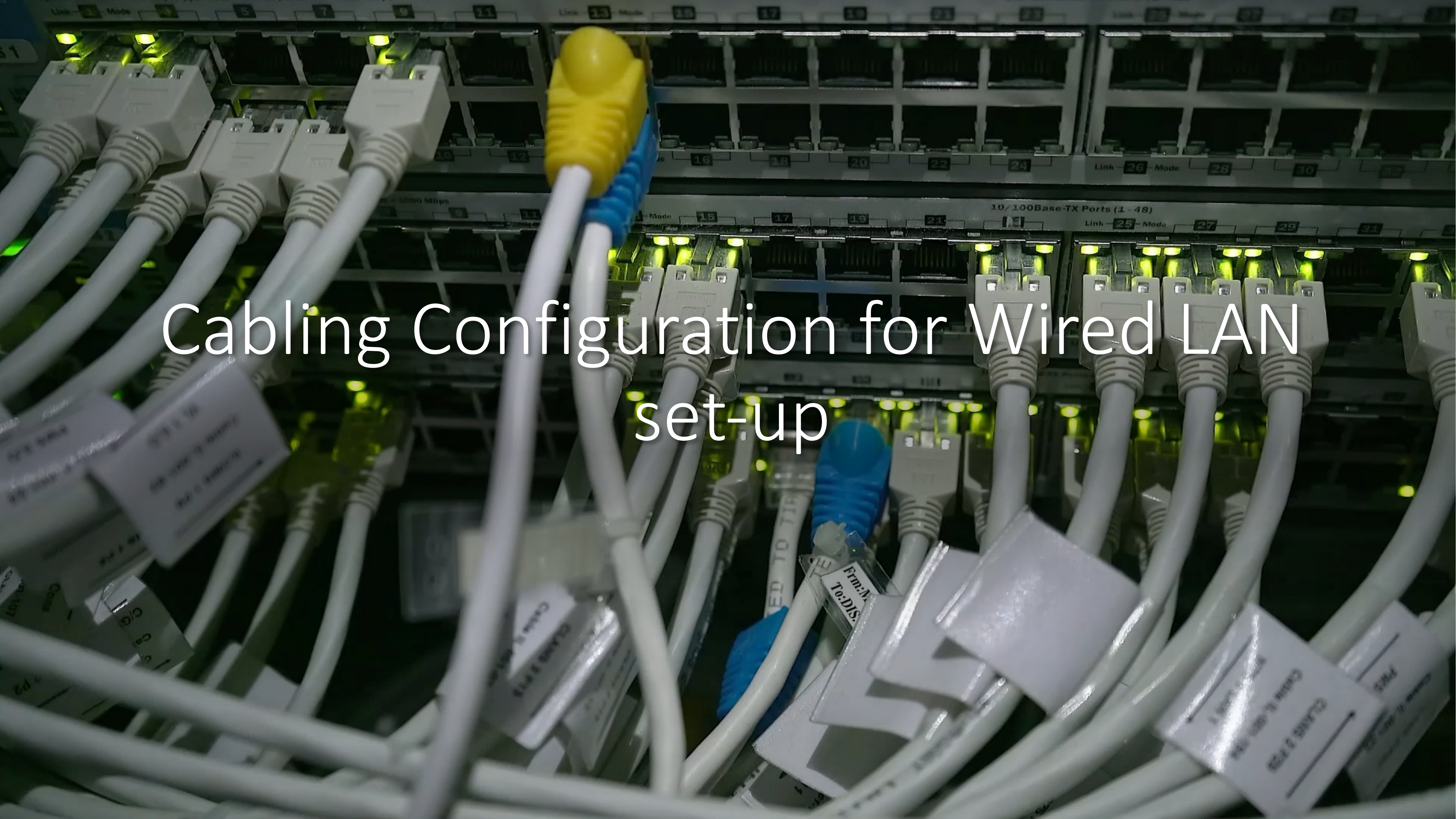
Network Cabling Configurations

Objectives:

- Identify the color configuration of crossover and straight-through cabling.
- Identify the tools/equipment needed for network cabling.
- Demonstrate how crossover and straight-through cabling is done.



Cabling Configuration for Wired LAN set-up

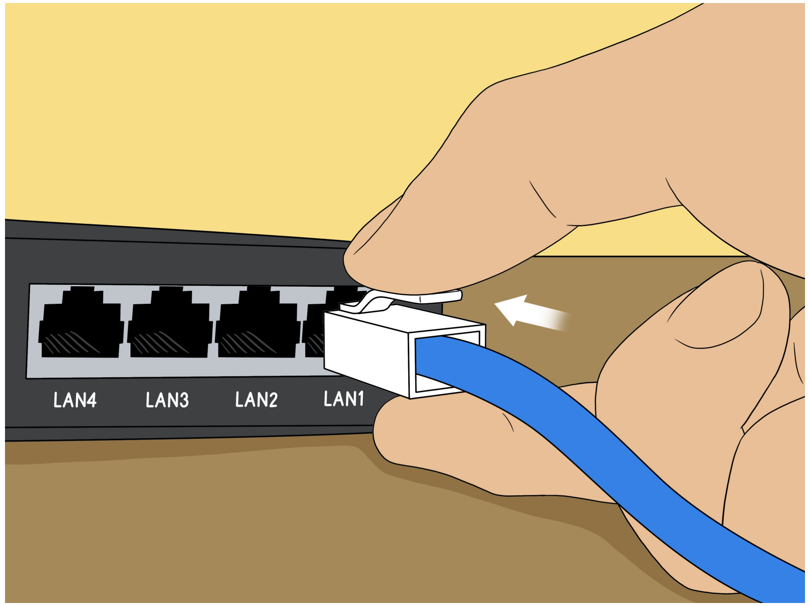
Materials Needed for Ethernet Cabling

- 2 pcs RJ 45
- UTB Cable
- Crimping Tool
- Tester

What is Ethernet Cable?

- An Ethernet cable is a network cable used for high-speed wired network connections between two devices.
- It is used for data transmission at both ends of the cable, which is called RJ45 connector.
- The Ethernet cables are categorized as Cat 5, Cat 5e, Cat 6, and UTP cable.
- Cat 5 cable can support a 10/100 Mbps Ethernet network while Cat 5e and Cat 6 cable to support Ethernet network running at 10/100/1000 Mbps.

Ethernet Cables



What is a crimping tool?

- A crimp tool is a tool that is used to join or connect two pieces of material by using compression to form a connecting bond.
- This type of hand tool is set to the desired pressure needed to effectively join two subjects together for a secure seal that allows a connection is made.
- Some crimp tools have pre-set dies for the terminal to ensure a precise seal. You can learn more in our complete

What is a cable tester?

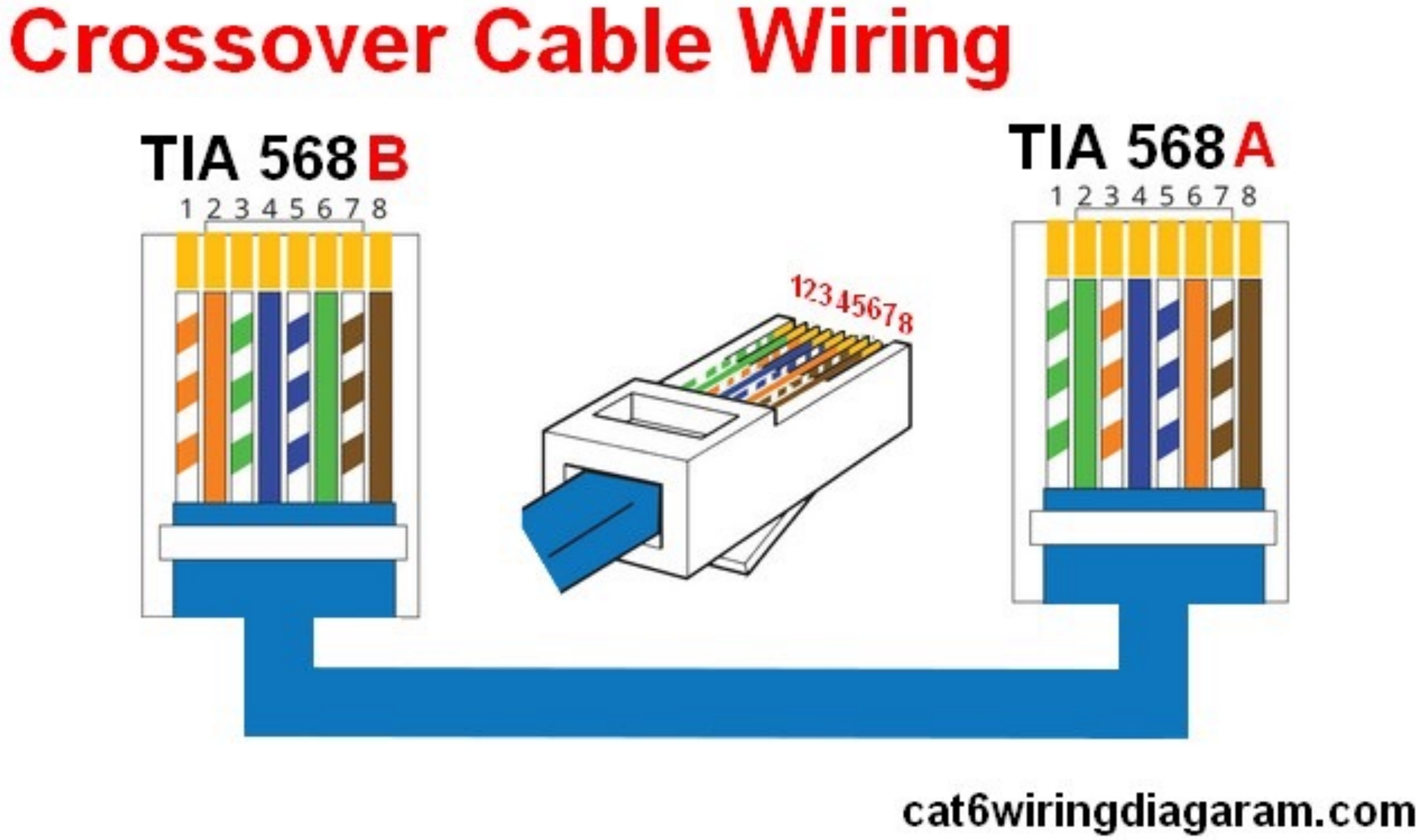
- A **cable tester** is a device used to test the strength and connectivity of a particular type of cable or other wired assemblies. There are many different types of cable testers.
- For computers, one of the most common types of cable testers used is for testing Cat 5, Cat 5e, and Cat 6 network cables. Because so many different types of data are transmitted over a network cable, a proper connection needs to be established between the computer and server.

Cable Testers

What is RJ45?

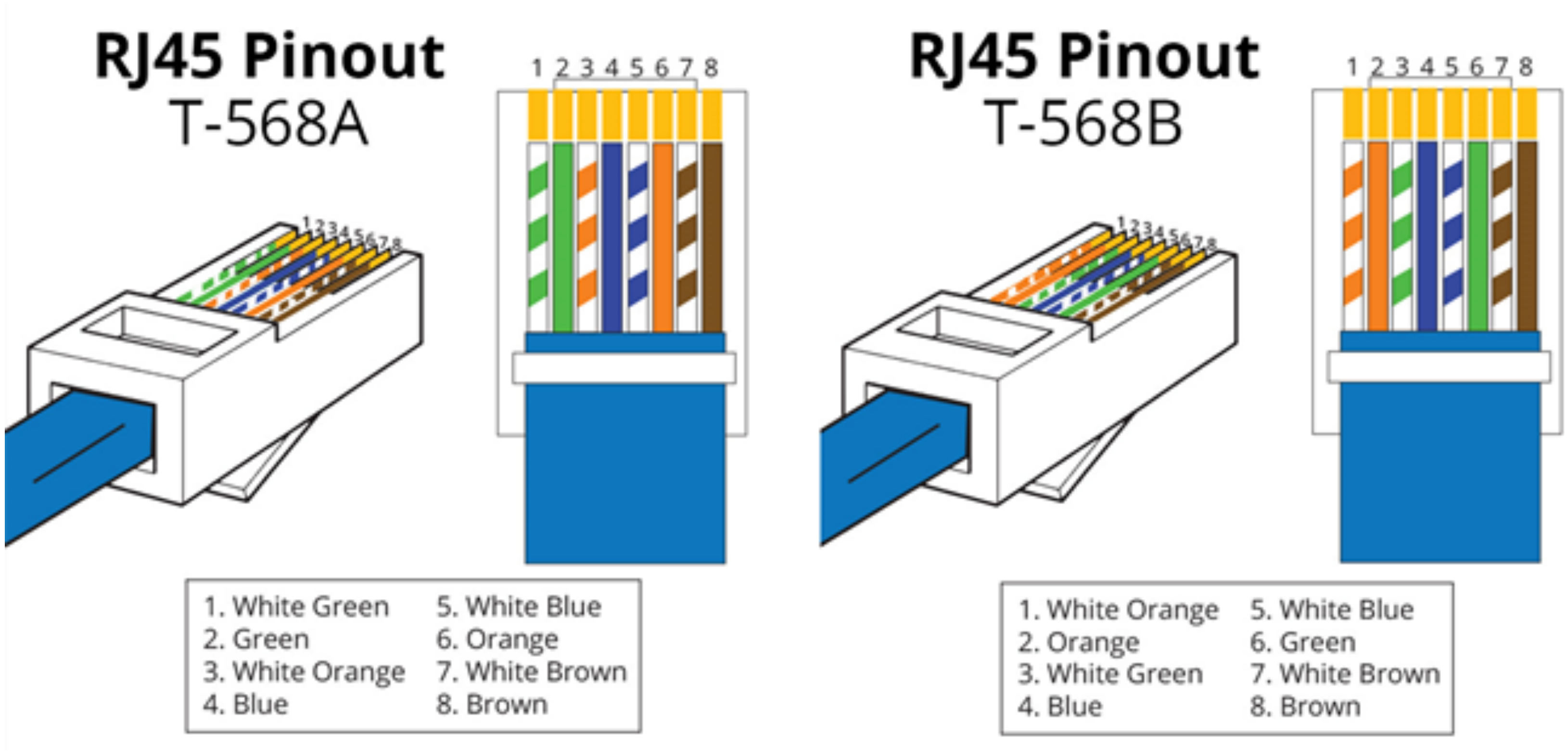
- A registered jack (RJ) is a standardized physical network interface for connecting telecommunications or data equipment.
- An 8-pin/8-position plug or jack is commonly used to connect computers onto Ethernet-based local area networks (LAN).
- Two wiring schemes–T568A and T568B–are used to terminate the twisted-pair cable onto the connector interface.

What is RJ 45?


Crossover Cable Wiring
TIA 568 B
1 2 3 4 5 6 7 8
TIA 568 A
1 2 3 4 5 6 7 8
12345678
cat6wiringdiagaram.com

How to configure network cabling

Crossover and Straight-through Cabling Configuration

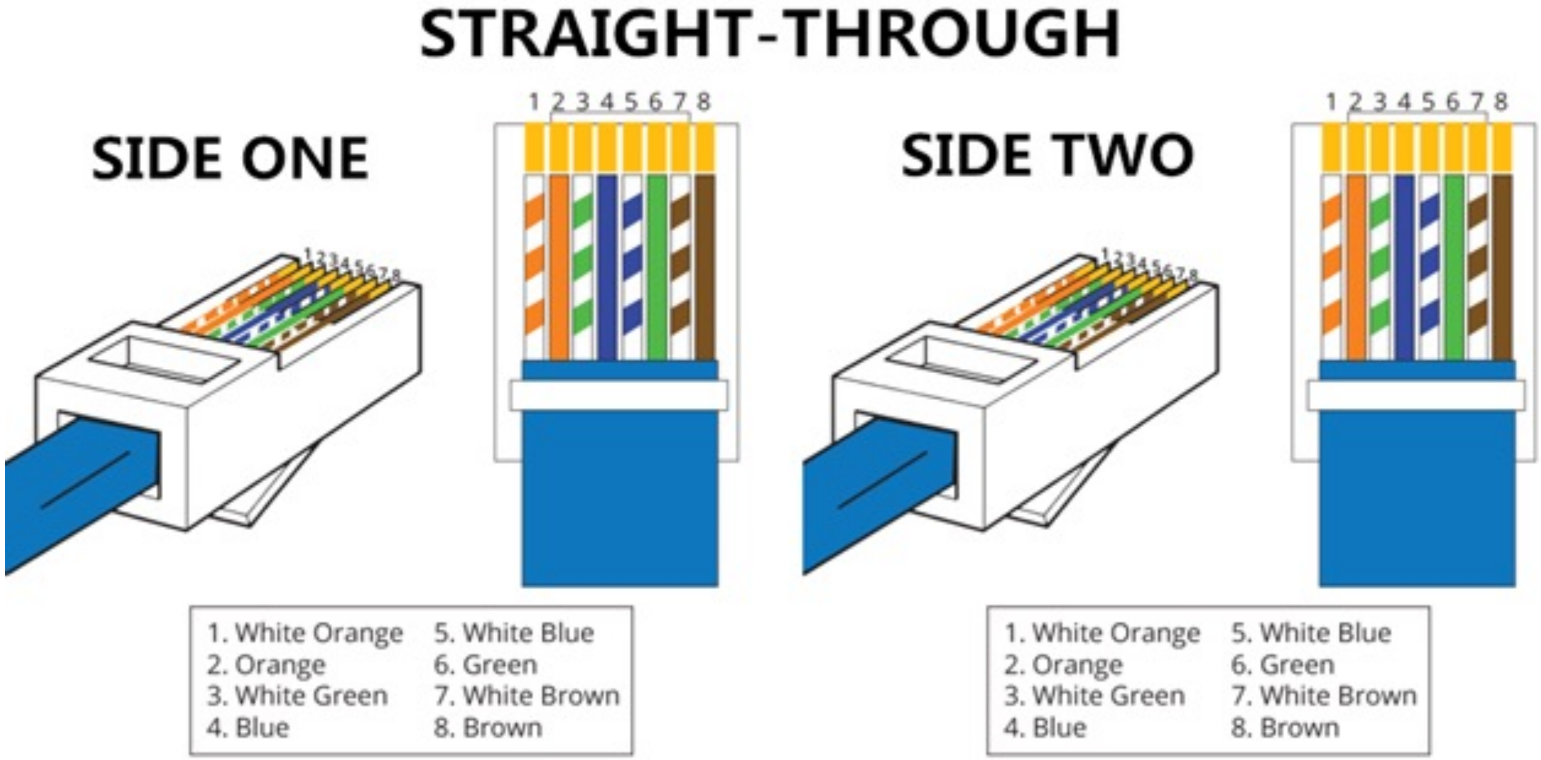


What is straight through?

- A straight through cable is a type of twisted pair cable that is used in local area networks to connect a computer to a network hub such as a router.
- This type of cable is also sometimes called a patch cable and is an alternative to wireless connections where one or more computers access a router through a wireless signal.
- On a straight through cable, the wired pins match. Straight through cable use one wiring standard: both ends use T568A wiring standard or both ends use T568B wiring standard.

What is straight through?

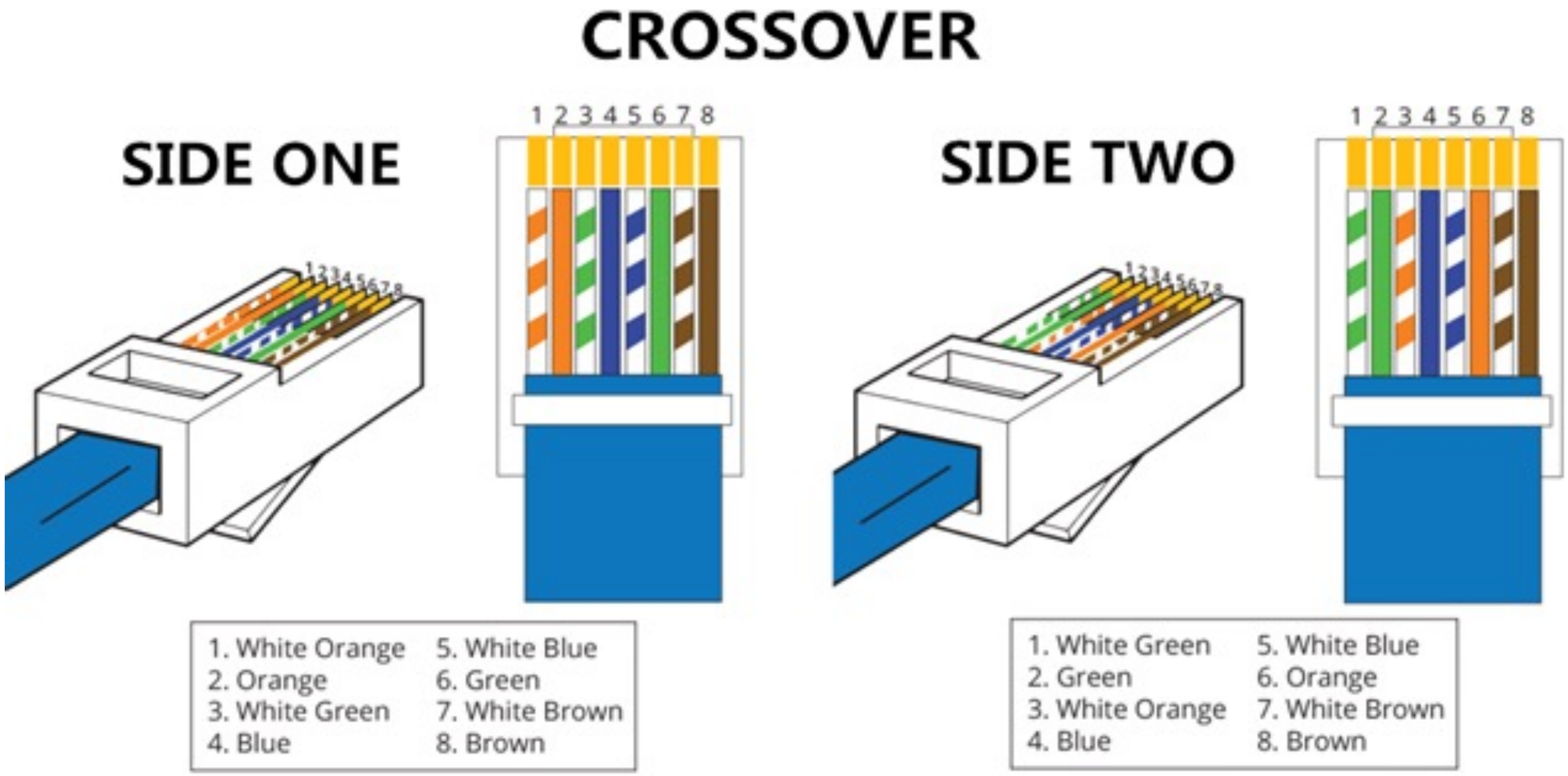
- The following figure shows a straight through cable of which both ends are wired as the T568B standard.



What is straight crossover cable?

- A crossover Ethernet cable is a type of Ethernet cable used to connect computing devices together directly.
- Unlike straight through cable, the RJ45 crossover cable uses two different wiring standards: one end uses the T568A wiring standard, and the other end uses the T568B wiring standard.
- The internal wiring of Ethernet crossover cables reverses the transmit and receive signals. It is most often used to connect two devices of the same type: e.g. two computers (via network interface controller) or two switches to each other.

What is straight crossover cable?



Crossover and Straight through

Helpful Video Links:

- Crossover Cabling:
- https://www.youtube.com/watch?v=LKxzJk08Ma8
- When to use crossover and straight-through:
- https://www.youtube.com/watch?v=ZrG4yqmRJUw